

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Winter 2023

Course: B. Tech. Branch: Civil Engineering and Allied Branches Semester: IV

Subject Code & Name: BTCVC406/BTCIC406/BTCEC406 Engineering Geology

Max Marks: 60

Date: 29/01/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|--------------|----------|
| A) Write short notes on: - | (CO1) | 6 |
| a. Dendritic drainage pattern | | |
| b. Sand dune | | |
| c. Weathering | | |
| B) Write a note on interior of the earth with neat labelled diagram. | (CO1) | 6 |
| C) Give detailed note on types of mountains and their origin with suitable examples. | (CO1) | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|---|--------------|----------|
| A) Write a short note on: - | (CO2) | 6 |
| a. Marble | | |
| b. Conglomerate | | |
| c. Metamorphism | | |
| B) What are the physical properties of minerals useful for the identification of minerals? | (CO2) | 6 |
| C) Write the classification of Igneous rocks based on depth, silica and texture. | (CO2) | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|--|--------------|----------|
| A) Write a short note on: - | (CO3) | 6 |
| a. Joints | | |
| b. Aquifer | | |
| c. Dyke | | |
| B) Define the term “unconformity” and describe various types of unconformities. | (CO1) | 6 |
| C) What are the geological faults and their various types? | (CO3) | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|--|--------------|----------|
| A) Write a short note on: - | (CO1) | 6 |
| a. RQD | | |
| b. Sampling technique | | |
| c. Surface Investigations | | |
| B) Explain the complete procedure of engineering geological investigation. | (CO3) | 6 |
| C) Write a detailed note on various drilling techniques used for the collection of samples. | (CO2) | 6 |

Q. 5 Solve Any Two of the following.

- | | | |
|---|--------------|----------|
| A) Write a short note on: - | (CO4) | 6 |
| a. Gravity dam | | |
| b. Suspension bridge | | |
| c. Tunnel Boring Machine | | |
| B) What are the major geological factors affecting the selection site for the dam? | (CO3) | 6 |
| C) Explain the various favourable and unfavourable geological conditions for reservoir site. | (CO4) | 6 |

***** End *****